

Computer Networks I

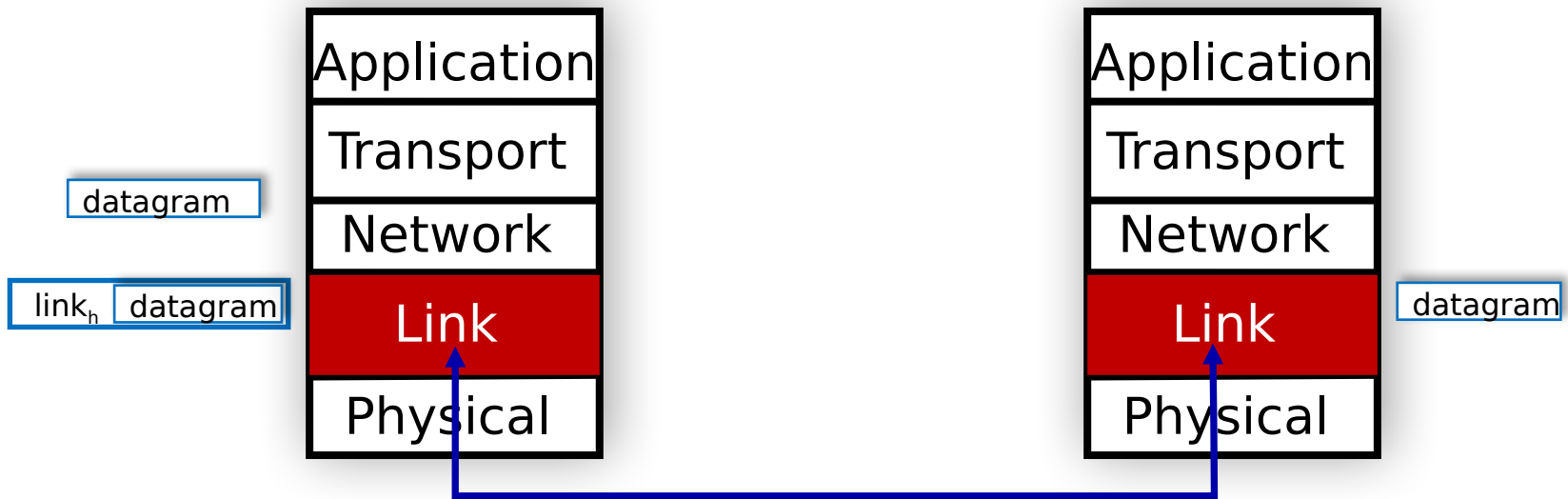
Data Link Control Protocols (Flow Control)

Amitangshu Pal
Computer Science and Engineering
IIT Kanpur

Data Link Layer

- Data link layer has responsibility of transferring datagram from one node to **physically adjacent** node over a link
 - Framing
 - Encapsulate datagram into frame, adding header, trailer
 - Reliable delivery between adjacent nodes
 - Flow control and error control
 - Channel access if shared medium
-

Data Link Layer



Sending side:

- Encapsulates datagram in frame
- Adds error checking bits, reliable data transfer, flow control, etc.

Receiving side:

- Looks for errors, reliable data transfer, flow control, etc.
- Extracts datagram, passes to upper layer at receiving side

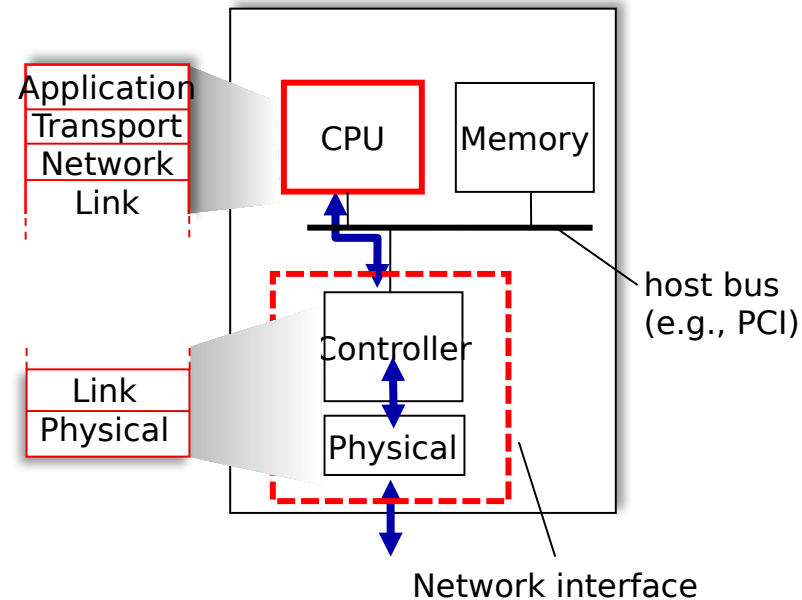
Data Link Layer



❑ Link layer implemented on-chip or in network interface card (NIC)

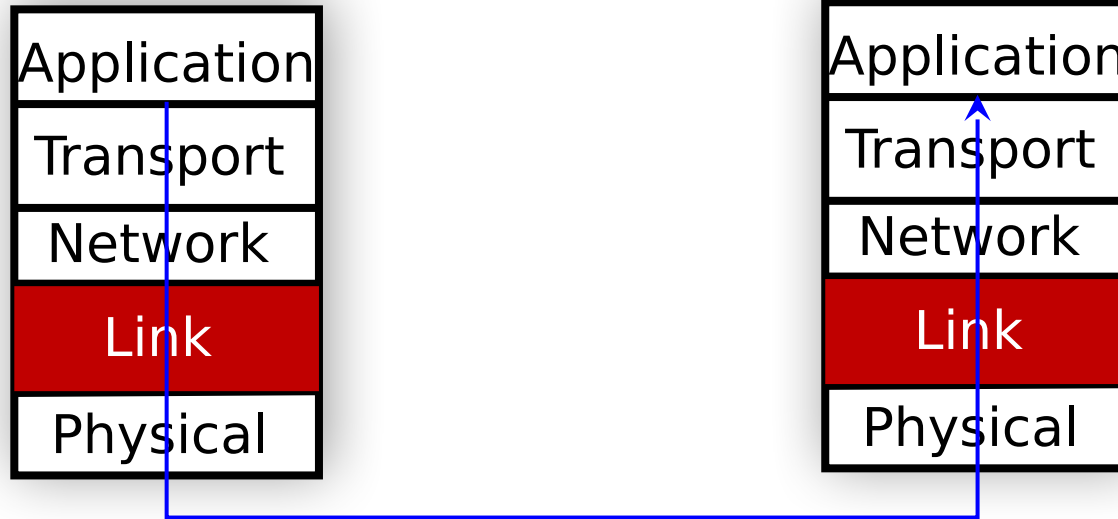
- Implements link, physical layer
- Attaches into host's system buses

❑ Combination of hardware, software, firmware



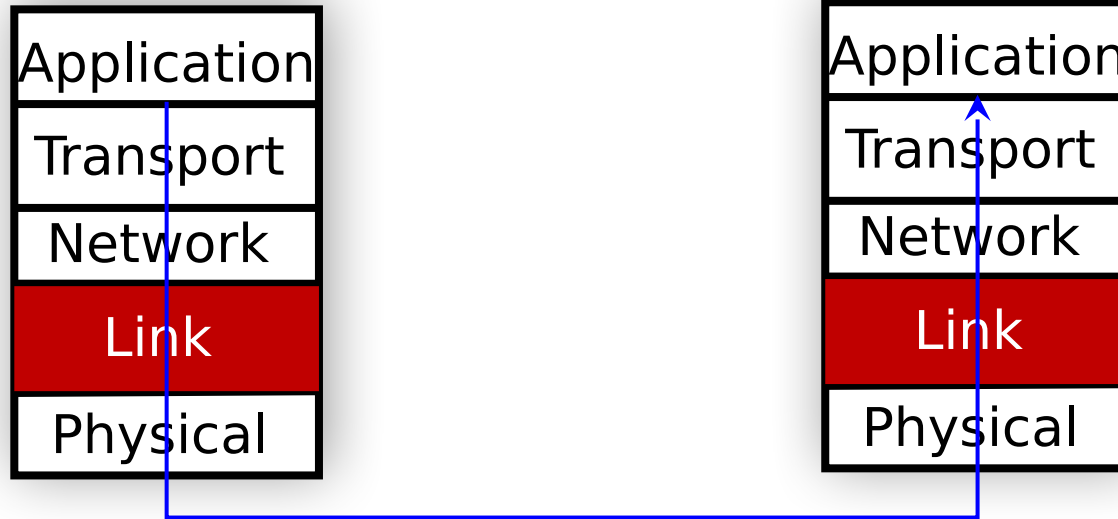
Flow Control

Flow Control



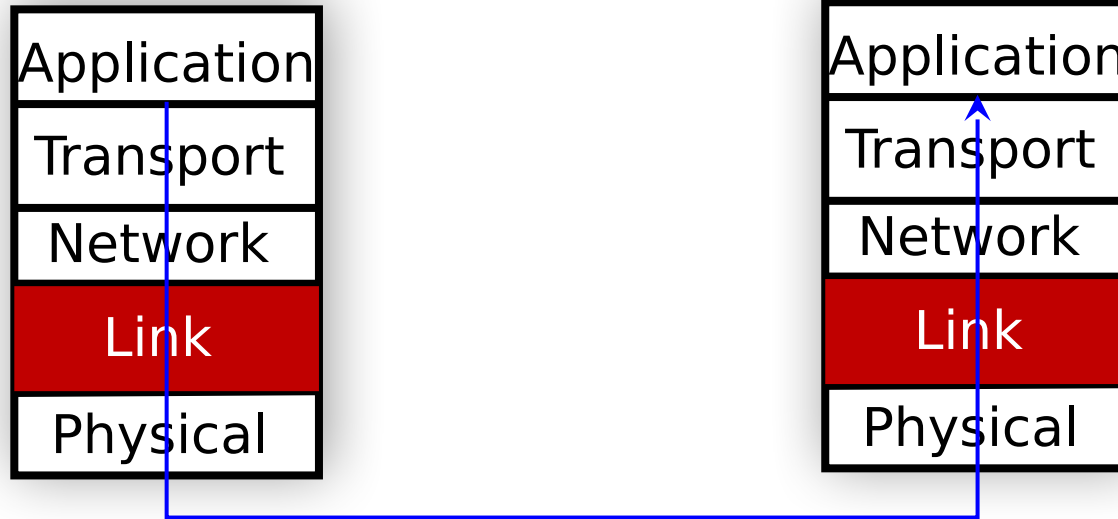
- Flow control assures that a transmitting entity does not overwhelm a receiving entity with data
 - If the transmitter transmits too fast, then the receiver buffer may overflow
 - This may lead to data loss, retransmission, and performance reduction

Flow Control



- Flow control assures that a transmitting entity does not overwhelm a receiving entity with data
 - **Assumptions:**
 - No frames loss or error, all transmitted frames are successfully received
 - Transmitted frames suffers from an arbitrary and variable amount of delay
-

Flow Control



□ Flow control:

- Stop and Wait Flow Control
 - Sliding Window Flow Control
-

Stop-and-Wait Flow Control

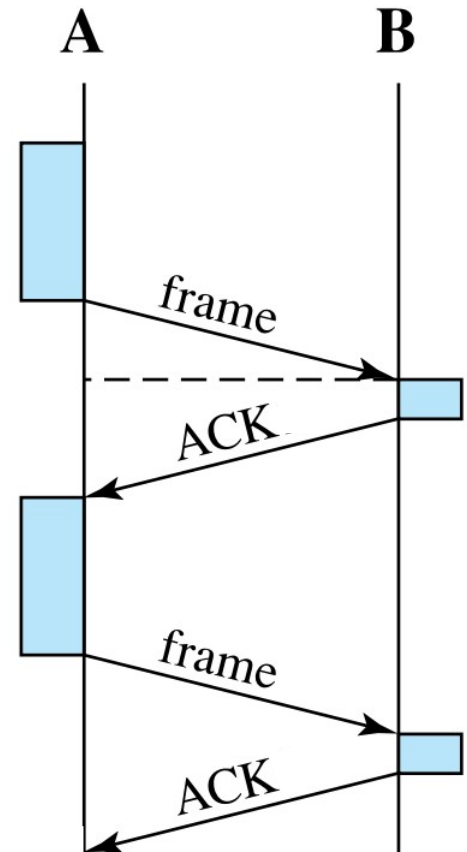
Stop and Wait Flow Control

□ Source:

- Source transmits a DATA frame
- Source waits for ACK frame before sending next DATA frame

□ Destination:

- Destination receives DATA frame → replies back with an ACK if ready to accept more data
- Destination can stop flow of data by withholding ACK

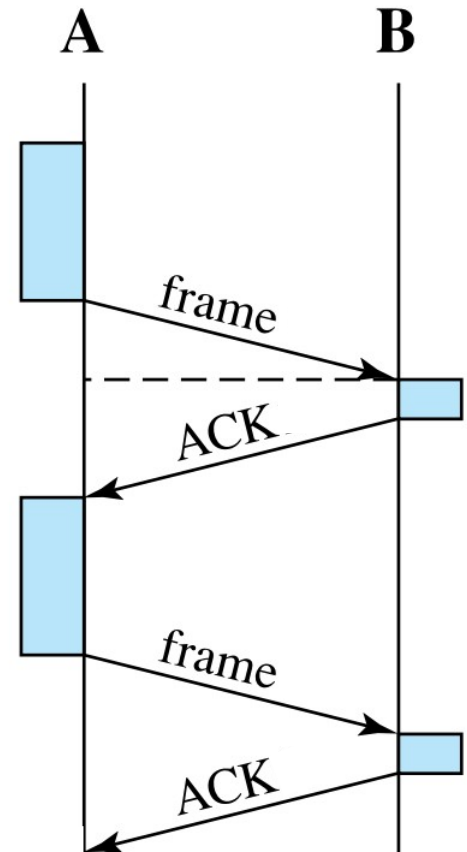


Efficiency Calculation

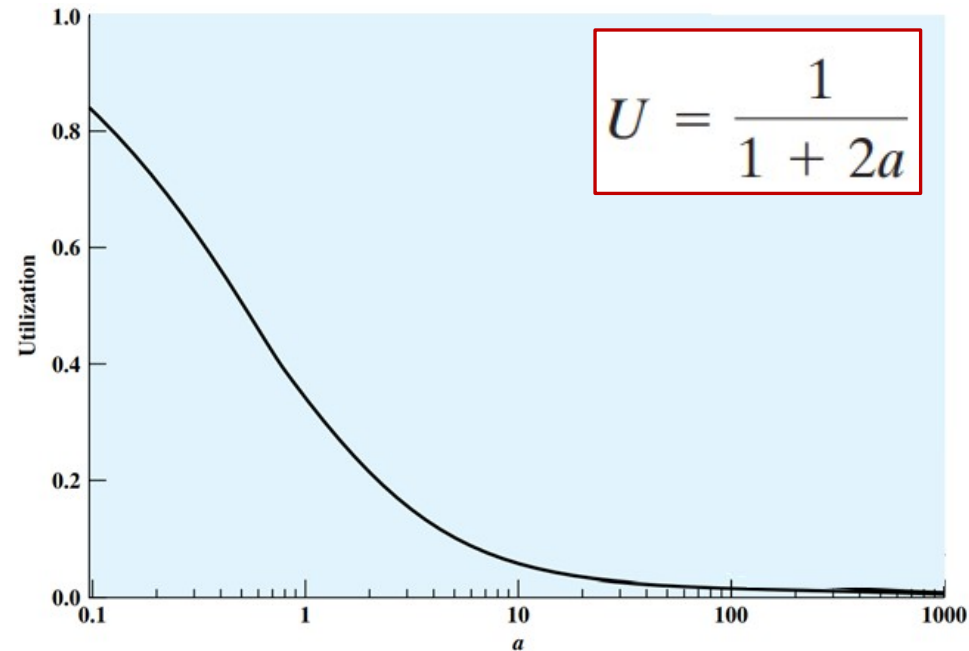
$$U = \frac{t_{\text{frame}}}{t_{\text{frame}} + t_{\text{ack}} + 2 \times t_{\text{prop}}}$$

$$a = \frac{\text{Propagation time}}{\text{Transmission time}} = \frac{t_{\text{prop}}}{t_{\text{frame}}}$$

$$U = \frac{1}{1 + 2a}$$



Efficiency Calculation



- ❑ Stop-and-wait allows only 1 frame to be in transit at a time
 - Leads to serious inefficiencies especially when the frame size is small
 - Efficiency can be greatly improved by allowing **multiple frames** to be in transit at a time → which is the idea of **Sliding-window flow control**

Sliding Window Flow Control

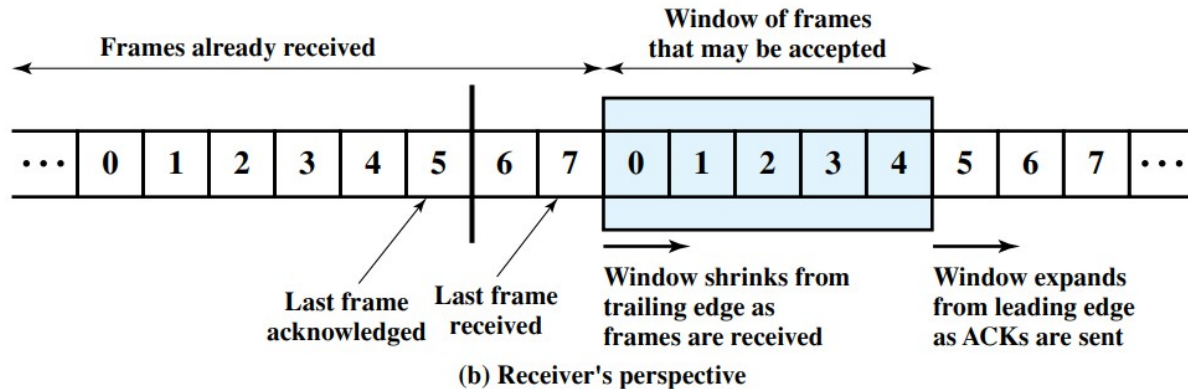
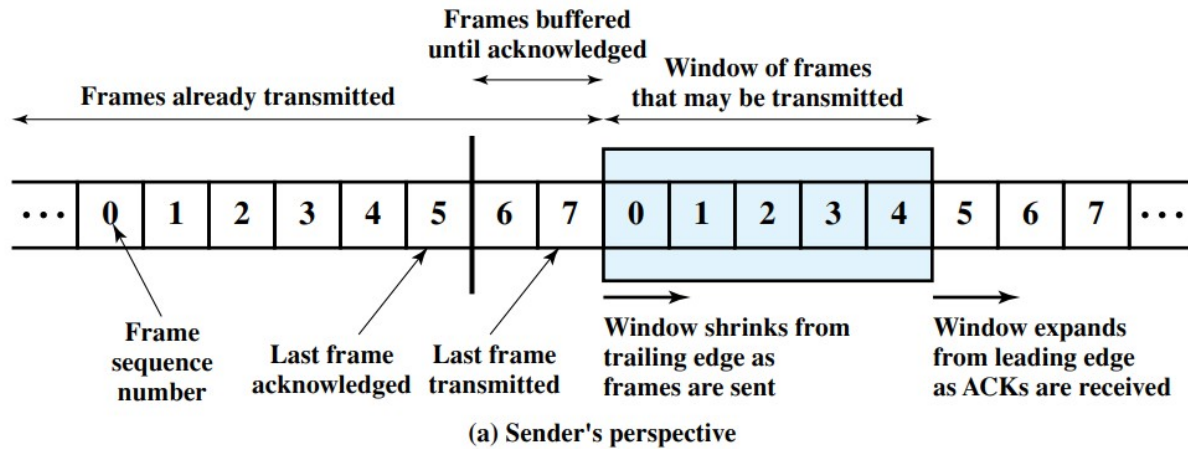
Sliding Window Flow Control

- ❑ Receiver allocates a buffer space of W frames
 - ❑ Sender can send up to W frames without waiting for an ACK

 - ❑ Each frame is labeled with a k -bit sequence number
 - Frames are numbered modulo 2^k
 - Giving max window size of up to $2^k - 1$ (will learn during error control)

 - ❑ Receiver acknowledges a frame by sending an ACK (or Receiver Ready, RR)
 - ❑ ACK includes the sequence number of the next frame expected
 - ACK also implicitly says that the receiver can receive the next W frames
-

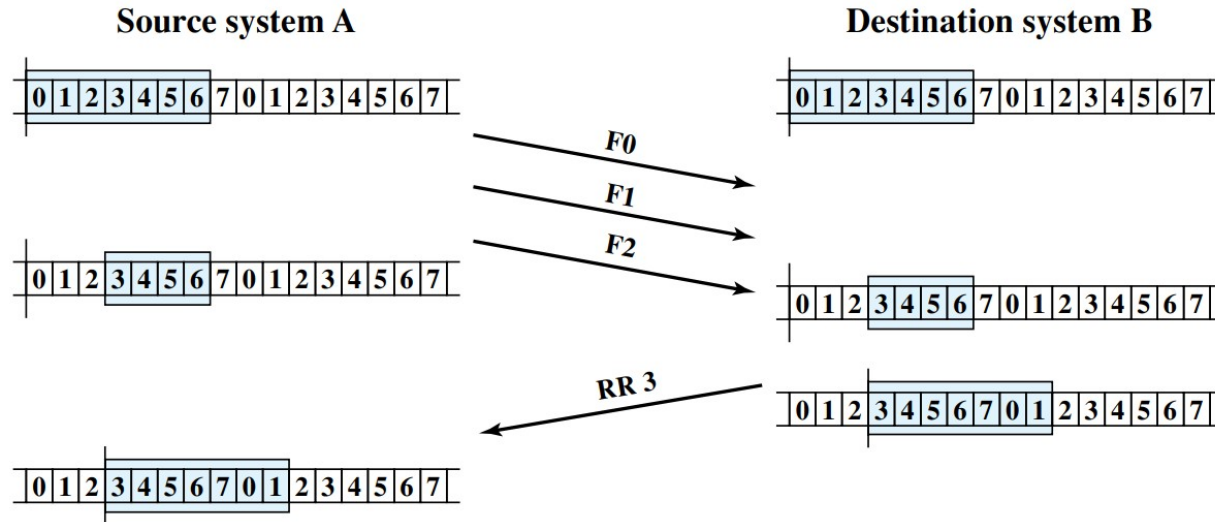
Sliding Window Flow Control



Sliding Window Flow Control

- Sender and Receiver record:
 - Last frame acknowledged
 - Last frame transmitted (in case of sender) or received (in case of receiver)
 - Window of frames that may be transmitted (in case of sender) or received (in case of receiver)
-

Sliding Window Flow Control: Example



Sliding Window Flow Control

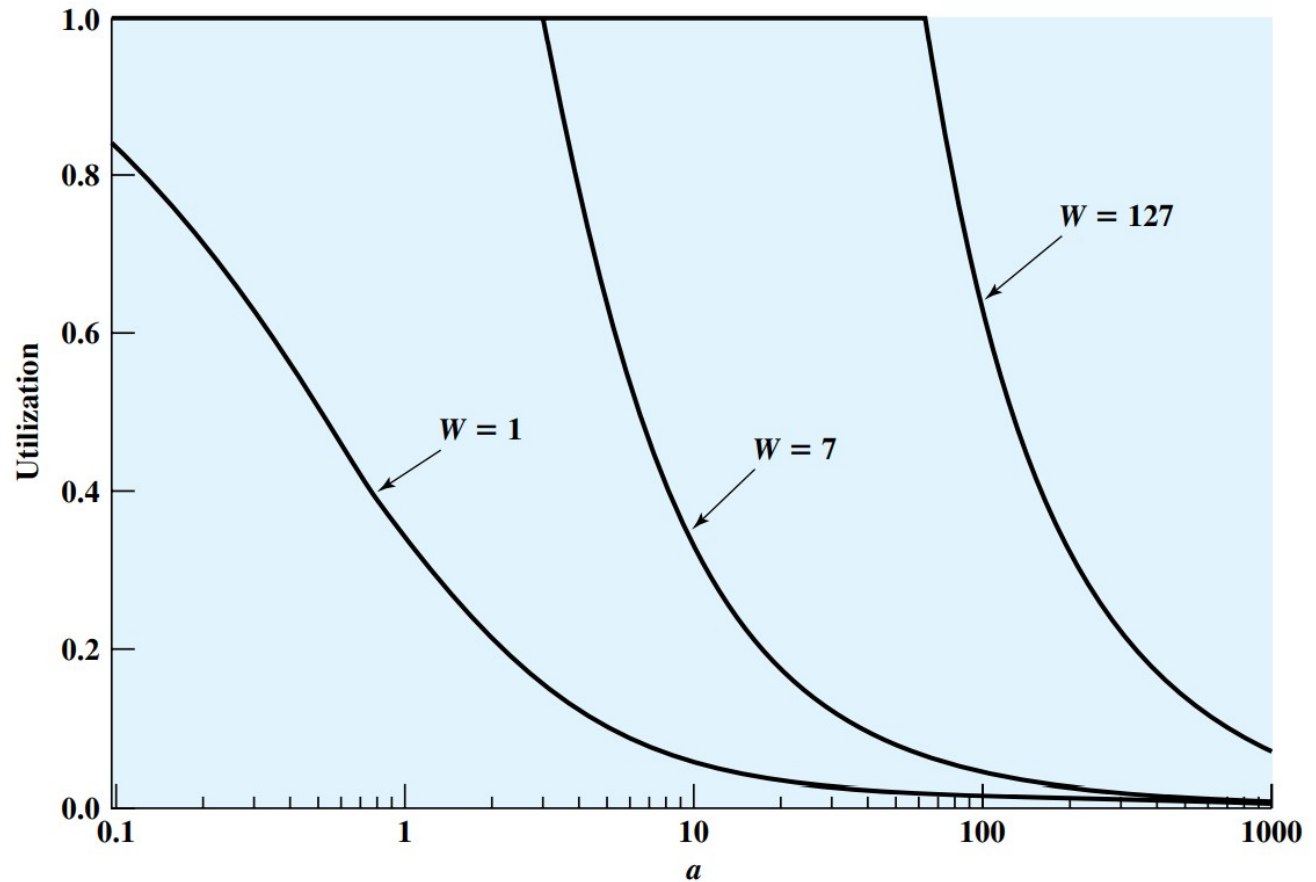
□ Additional features:

- **Receive Not Ready (RNR):** Receiver can ACK frames without permitting further transmission
 - RNR 5: received frames up to 4, but unable to accept any more
 - Must send a normal acknowledge to resume

 - **Piggybacking:** If a station has a DATA and ACK to send, it sends both together → **saves communication capacity**
 - DATA frame includes sequence number of the frame and sequence number used for ACK (i.e. next expected frame number)
 - If the station has an ACK but no DATA to send, it sends normal ACK (RR or RNR)
 - If the station has DATA to send but no new ACK → repeats the last ACK seq. no.
-

Sliding Window Flow Control

$$U = \begin{cases} 1 & W \geq 2a + 1 \\ \frac{W}{2a + 1} & W < 2a + 1 \end{cases}$$



Summary

❑ Link layer services

❑ Flow control in Link Layer:

- Stop and wait
 - Sliding window
-